

RESEARCH REPORT

Grip strength comparison in immune-mediated neuropathies: Vigorimeter vs. Jamar

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Abstract The Jamar dynamometer and Vigorimeter have been used to assess grip strength in immune-mediated neuropathies, but have never been compared to each other. Therefore, we performed a comparison study between these two devices in patients with immune-mediated neuropathies. Grip strength data were collected in 102 cross-sectional stable and 163 longitudinal (new diagnoses or changing condition) patients with Guillain-Barré syndrome (GBS), chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), gammopathy-related polyneuropathy (MGUSP), and multifocal motor neuropathy (MMN). Stable patients were assessed twice (validity/reliability studies). Longitudinal patients were assessed 3–5 times during 1 year. Responsiveness comparison between the two tools was examined using combined anchor/distribution-based minimum clinically important difference (MCID) techniques. Patients were asked to indicate their preference for the Jamar or Vigorimeter. Both tools correlated highly with each other ($\rho = 0.86$, $p < 0.0001$) and showed good intra-class correlation coefficients (Jamar [Right/Left hands]: ICC 0.997/0.96; Vigori: ICC 0.95/0.98). Meaningful changes were comparable between the two instruments, being higher in GBS compared to CIDP patients. In MGUSP/MMN poor responsiveness was seen. Significant more patients preferred the Vigorimeter. In conclusion, validity, reliability, and responsiveness aspects were comparable between the Jamar dynamometer and Vigorimeter. However, based on patients' preference, the Vigorimeter is recommended in future studies in immune-mediated neuropathies.

Key words: grip strength, Jamar dynamometer, responsiveness comparison, Vigorimeter

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Introduction

Weakness and sensory deficits in patients with immune-mediated neuropathies have a major impact on activities of daily living, social functioning, and quality of life (Bernsen et al., 1997). Because distal weakness predominates in these conditions, grip strength has been used to assess outcome and evaluate treatment effects in patients with hand weakness (Rhind et al., 1980; Sunderland et al., 1989; Merkies et al., 2000; Krischak et al., 2009). The Jamar dynamometer and the Vigorimeter are the most commonly used tools to assess grip strength in immune-mediated neuropathy trials (Hahn et al., 1996; Federico et al., 2000; Hughes et al., 2001; Hughes et al., 2008). Both have showed good scientific properties (Mathiowetz et al., 1984; Solgaard et al., 1984; Jones et al., 1991; Bellace et al., 2000; Merkies et al., 2000; Rosen et al., 2000; Mathiowetz, 2002; Smidt et al., 2002; Bohannon and Schaubert, 2005; Guerra and Amaral, 2009; Lindstrom-Hazel et al., 2009; Espana-Romero et al. 2010), although a clinimetric comparison between these measures has never been performed in immune-mediated neuropathies (Merkies and Lauria, 2006). If one of these devices is a superior measure of handgrip strength, it could be used as a standard for future clinical trials, thereby increasing precision, reliability, and responsiveness within a treatment trial, and allowing comparability between therapeutic trials. Therefore, the purpose of this study is to compare the validity, reliability, and responsiveness of both tools and select the superior device as the standard grip strength measure for future studies in patients with immune-mediated neuropathies.

Methods

Patients

Cross-sectional patients

Between April 2007 and January 2010, 102 patients with clinically stable immune-mediated neuropathies participated in this study: $n=30$ patients with Guillain Barré syndrome (GBS), $n=30$ with chronic inflammatory demyelinating polyradiculoneuropathy (CIDP), $n=20$ with IgM anti-MAG+ gammopathy-related polyneuropathy (MGUSP), and $n=22$ with multifocal motor neuropathy (MMN).

Longitudinal patients

Between August 2007 and December 2012, a total of 163 patients with a newly diagnosed (or relapsing) immune-mediated neuropathy (GBS: $n=55$, CIDP: $n=59$, MGUSP: $n=23$, MMN: $n=26$) were enrolled and examined prospectively over 12 months. Patients

were recruited at 26 international neuromuscular centers as part of the Peripheral Neuropathy Outcome Measures Standardization (PeriNomS) study (van Nes et al., 2008; Nes, 2012).

Eligibility

All cross-sectional patients had residual symptoms and signs and a stable neurological and functional status over the 2 months prior to enrollment (with or without interval treatment), which was confirmed by patient's interview or record's review. All longitudinal patients were newly diagnosed or had a relapse after at least 2 months without treatment and were treated to the discretion of their own physician.

Recruitment location and ethical approval

Cross-sectional patients were recruited at University outpatient clinics in the Netherlands (Rotterdam, Utrecht, and Maastricht). Longitudinal patients were recruited during hospital admission or at outpatient clinics in the United States (Boston and Detroit), Canada (Ontario and Toronto), Brazil (Ribeirão Preto), UK (London), Italy (Milan, Rome, and Venice), France (Paris and Marseille), Spain (Barcelona), Belgium (Brussels), and The Netherlands (Maastricht, Amsterdam, Utrecht, and Rotterdam). The local Medical Ethics Committee in each participating center approved the protocol. Written informed consent was obtained from all participants.

Grip strength assessment

Grip strength assessment was performed using the Jamar hydraulic hand dynamometer (Sammons Preston Rolyan, Bolingbrook, IL, USA) and the Vigorimeter (Martin, Tuttlingen, Germany). The Jamar dynamometer quantifies isometric force in Pounds (range: 0–200). The Vigorimeter quantifies grip strength on a manometer after squeezing a rubber bulb, and is scored in Kilopascals (range: 0–160). All tools were newly purchased and provided to study investigators before study onset. Position two of the adjustable handle of the Jamar dynamometer and the middle size bulb (diameter 4.8 cm) of the Vigorimeter were used. Patients were examined according to the recommendations by the American Society of Hand Therapists (Casanova, 1992). Three maximum voluntary contractions were recorded for each hand in alternating order (resting interval period: ~30 s). The mean values obtained from both hands were used for analyses. At each visit patients were asked if they had a preference for the Jamar dynamometer or the Vigorimeter (*t* using a 5-point Likert scale: "great preference to the Vigorimeter," "slight preference for the Vigorimeter," "no preference for either tools", "slight preference for the Jamar", "great preference for the Jamar").

Study design

All researchers were trained how to perform and record the findings of the two grip-strength tools in a standardized manner. Assessment of the instruments was performed in a random order, twice in the cross-sectional group of patients at two visits (interval 2–4 weeks to determine test-retest reliability for each hand using each instrument; first assessments were used for validity correlation studies between the instruments). In the cross-sectional group of patients, all assessments were performed by two investigators (S. I. v. N./E. K. V.). Both researchers examined the patient at first examination, and one researcher (S. I. v. N.) examined the patient a second time (2–4 weeks interval). In the longitudinal group, various investigators examined the patients serially (for GBS/CIDP at T0, T1, T3, T6, and T12 months; for MGUSP/MMN at T0, T3, and T12 months).

Statistics

Validity and reliability studies

Validity and reliability studies were performed in the cross-sectional group of patients; these patients were examined by two researchers (S. I. v. N./E. K. V.) separately at first assessment (inter-observer studies), and by one researcher (S. I. v. N.) a second time (2–4 weeks) later (intra-observer studies). Responsiveness was determined in the longitudinal group of patients.

Correlations between the Vigorimeter and the Jamar were calculated (validity studies: Spearman's rank correlation coefficient). Furthermore, inter-/intra-reliability scores of both tools were compared by estimating the intra-class correlation coefficients using a one-way random-effects analysis of variance model (ANOVA; expressed as R^2).

Responsiveness studies

Responsiveness comparison between the two tools was examined using the concept of minimum clinically important difference (MCID) (Jaeschke et al., 1989; Sloan et al., 2003; Merkies et al., 2010). However, this concept has been criticized because different methods produce different MCID cut-off values (Beaton et al., 2002). It was therefore decided in the current study to adopt a stringent approach by combining an anchor-based and a distribution-based MCID method, as was recently suggested (Merkies et al., 2010). Anchor-based MCID methods compare changes in a scale's score with an external criterion (e.g., the patients' own judgment of their clinical status after therapy) and distribution-based methods are based on the statistical characteristics of the scores obtained (e.g., effect size) (Jaeschke et al., 1989; Beaton et al., 2002; Sloan et al., 2003). As previously assessed in

inflammatory neuropathy patients, the SF-36 question number 2 was chosen to determine global change in clinical status at each visit compared to 1 year ago (Ware et al., 2000). The anchor-based MCID was determined as the mean score difference between the response options "somewhat better now than 1 year ago" and "about the same as 1 year ago" for the records obtained at the first three assessments.

The universal benchmark rule of $1/2 \times$ standard deviation (SD) was chosen as the distribution-based method for the purposes of the current study (Norman et al., 2003; Sloan et al., 2003). The theoretical range for the grip strength for both instruments was based on available longitudinal data at entry.

The MCID cutoff values were determined for both grip strength instruments using the selected anchor- and distribution-based MCID techniques. For both instruments, a patient was defined as a responder if the grip strength value at assessment of the patient improved enough to reach both MCID thresholds for the instrument being used (coded 1: both MCID thresholds were reached; 0 if only one or none of the cut-offs were reached). The proportion of patients classified as responder was calculated and compared as previously described: we created Kaplan–Meier curves to estimate the cumulative proportion of patients being classified as being a responder for each instrument (overall and separately in GBS, CIDP, MGUSP, and MMN) over time (at 1/3/6/12 months of follow-up), using log-rank test for comparison (van der Beek et al., 2013). Finally, the preference of the patients for one instrument over the other was also compared (chi-squared test). All analyses were undertaken using Stata 12.0 for Windows. A p-value of <0.05 was considered significant.

Results

General characteristics

Patient characteristics at entry are presented in Table 1. Most individuals were right-handed (cross-sectional group: 82.3%; longitudinal group: 88.7%) and male. A total of $n=45$ patients of the cross-sectional were clinically stable with interval treatment: 23 patients with CIDP, 17 with MMN, and 1 with MGUSP were treated with periodic infusions of intravenous immunoglobulin (IVIg). Two CIDP patients received plasma exchange, one CIDP patient was treated with corticosteroids, and one MGUSP patient was treated with Rituximab. Most longitudinal patients received medical intervention (93% in GBS, 71.2% in CIDP, 50% in MGUSP, and 84% in MMN) as previously reported (Draak et al., 2014).

Table 1. Characteristics of study population

Illness	Cross-sectional population N = 102				Longitudinal population N = 163			
	GBS (30)	CIDP (30)	MGUSP (20)	MMN (22)	GBS (55)	CIDP (59)	MGUSP (23)	MMN (26)
Age, mean (SD), range	61.5 (9.2), 47–76	58.1 (10.8), 26–74	68 (7.9), 54–81	54 (14.3), 29–75	53.4 (17.7), 19–90	52.7 (14.7), 18–79	62.8 (10.6), 45–85	48.8 (11.8), 29–68
Gender, n (%)								
Female	17 (56.7)	9 (30)	3 (15)	5 (22.7)	18 (32.7)	17 (28.8)	8 (34.8)	5 (19.2)
Male	13 (43.3)	21 (70)	17 (85)	17 (77.3)	37 (67.3)	42 (71.2)	15 (65.2)	21 (80.8)
Years since diagnosis	14.3 (6.8), 5–32	11.1 (5.6), 4–26	14.1 (6.5), 6–27	18.8 (9.3), 9–48	–*	1.8 (3.1), 0–16	3.3 (5.3), 0.2–25	2.7 (2.9), 0–9.8
Mean (SD), range								
Grip strength, mean (both hands) (SD), range	65.4 (31.8), 1.2–123.3	64.6 (24.9), 17–120.3	73.1 (27.1), 24.3–117.5	32 (33.8), 0–117.8	26.3 (27.5), 0–99.2	42.3 (28.3), 0–115.7	68.4 (27.5), 30.8–125.8	57.2 (28.1), 10.3–102.5
Jamar (Pounds)	71.7 (27.7), 16.5–111.7	80.7 (31.5), 17.8–139.8	77 (31.2), 22.5–139	49.2 (35.3), 0–120.3	31.6 (29.3), 0–100.5	48.3 (28), 0–108.3	73.1 (22.2), 37.8–119.2	63.8 (27.3), 9.7–120.7
Vigrometer (KiloPascals)								

CIDP, chronic inflammatory demyelinating polyradiculoneuropathy; GBS, Guillain-Barré syndrome; MGUSP, gammopathy-related polyneuropathy; MMN, multifocal motor neuropathy; SD, standard deviation. *GBS is a sub-acute illness and all patients with GBS in the longitudinal group were newly diagnosed.

Validity and reliability studies

Data obtained from the cross-sectional group of patients (n=102) were used for these purposes. There was great equivalence between the right and left hand findings, as has been previously observed (Merkies et al., 2000, 2010). Therefore, an overall mean score for both hands was used for further clinimetric comparison. A significant correlation was found between the two grip strength tools (Spearman's rho 0.86, $p < 0.0001$, n=102 patients examined). High comparable intra-class correlation coefficients were found for both tools (Jamar, intra-/inter-observer scores: 0.99 [n=101]/0.96 [n=99]; Vigorimeter, intra-/inter-observer scores: 0.95 [n=100]/0.98 [n=98]).

Responsiveness studies

Data from 163 serially examined patients were used for these studies. The calculated MCID_{anchor} was six for both instruments. The calculated MCID_{distribution} threshold for the Vigorimeter was 10 and for the Jamar 10.5. Overall, up to 70% of the patients showed a cumulative meaningful improvement (thus reaching both MCID_{anchor} + MCID_{distribution} cut-offs) for both instruments. The findings were slightly higher for the Vigorimeter, but the difference with the Jamar was non-significant (Figure 1A). At the disease level, significantly higher cumulative meaningful and comparable improvement was seen for both instruments in patients with GBS when compared to the other illnesses ($p < 0.0001$ for both instruments) (Figures 1B and 1C). For the patients with MGUSP and MMN, the responsiveness findings were poor for both instruments (Figures 1B and 1C); however, in MMN the proportion of responsiveness was significantly higher using the Vigorimeter when compared with the Jamar dynamometer findings ($p = 0.0074$). In MGUSP, the findings for both instruments were not different.

Patients' preference

Comparing patients' preference for one of the instruments demonstrated a higher proportion in favor of the Vigorimeter over the Jamar (cross-sectional group: 52.2%: 29%, p -value: 0.033; longitudinal: 53%: 25.3%, p -value: 0.014).

Discussion

We found that grip strength assessment with the Jamar dynamometer and the Vigorimeter demonstrated comparable validity and reliability, as suggested in prior studies (Mathiowetz et al., 1984; Solgaard et al., 1984; Jones et al., 1991; Bellace et al., 2000; Merkies et al., 2000; Rosen et al., 2000; Mathiowetz,

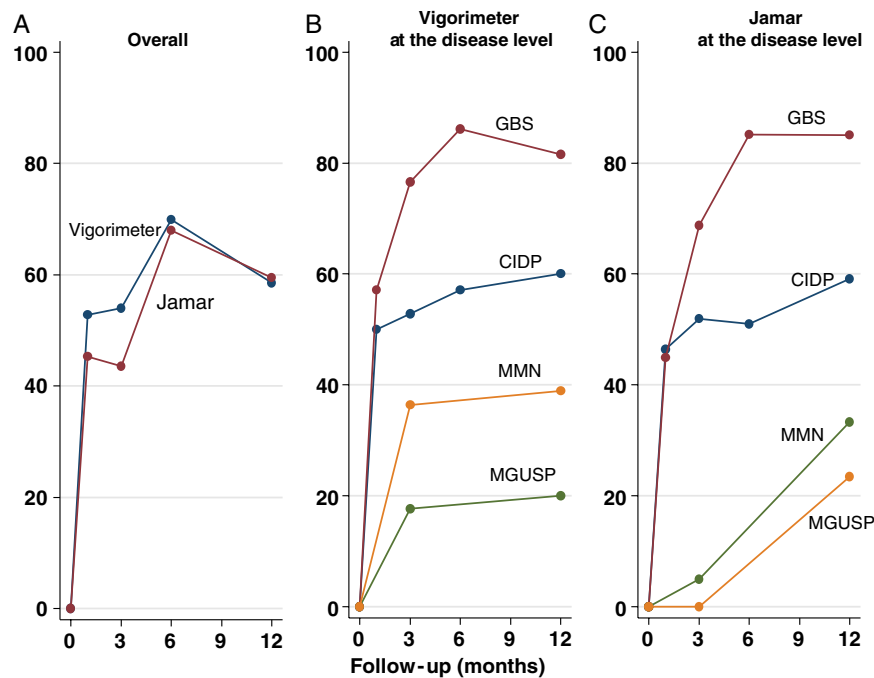


Figure 1. Significant grip strength improvement in immune-mediated neuropathies. GBS: Guillain-Barré Syndrome, CIDP: chronic inflammatory demyelinating polyradiculoneuropathy, MGUSP: monoclonal gammopathy of undetermined significance related polyneuropathy; MMN: multifocal motor neuropathy. A high proportion of patients reaching meaningful improvement (thus reaching $MCID_{\text{anchor}} + MCID_{\text{distribution}}$ predefined cut-offs) was seen for both instruments. The findings were consistently higher in GBS > CIDP; for MGUSP and MMN, the responsiveness findings were poor.

2002; Smidt et al., 2002; Bohannon and Schaubert, 2005; Guerra and Amaral, 2009; Lindstrom-Hazel et al., 2009; Espana-Romero et al. 2010). Only one other study, in healthy children, directly compared reliability aspects of the Vigorimeter and a Jamar-like tool (Lode dynamometer) and showed significantly lower intra-class correlation coefficients for the Vigorimeter (Molenaar et al., 2008). In contrast, we observed equally high intra-class correlation coefficients for both devices. These findings suggest that both tools are equally valid and reliable in immune-mediated neuropathies. Responsiveness findings for both devices were also comparable and non-significantly different (Figures 1A and 1C), were systematically higher in patients with GBS > CIDP, and were poor in MGUSP/MMN subpopulation. Similar differences were also seen for a recently developed inflammatory activity and participation scale, called the I-RODS (van Nes et al., 2011; Draak et al., 2014).

Our study demonstrated patients prefer the Vigorimeter. Patients arguments for preferring the Vigorimeter over the Jamar dynamometer include, “less heavy”, “easier”, “more comfortable”, “less painful to squeeze a bulb”, and “squeezing makes you feel the force you are giving”. Those who preferred the Jamar dynamometer listed arguments such as “more robust”, “more suitable for big hands,” and “less

chance of using it incorrectly.” The Vigorimeter showed to capture both meaningful improvement and deterioration earlier than the widely used disability INCAT scale (Hughes et al., 2008; Vanhoutte et al., 2013b). This is particularly of importance for improvement as well as maintenance trial designs. In addition, age and gender dependent normative values have also been published for these instruments that can be used in the clinical setting (Mathiowetz et al., 1984; Merckies et al., 2000; Werle et al., 2009). Based on our observations indicating equivalent validity, reliability, and responsiveness, it seems that the Vigorimeter should be selected as the standard device for grip strength assessment in future studies based upon patient’s preference. In a recent consensus meeting the Vigorimeter was proposed as one of the cardinal assessment tools in addition to the I-RODS (Vanhoutte et al., 2013a).

There are some methodological issues that should be addressed. The number of serially recruited patients with MGUSP and MMN were low. In addition, there was no standardized treatment for patients with MGUSP. These observations may have influenced the responsiveness results obtained. It is conceivable that a longer period of observation, preferably >1 year, is needed in these indolent illnesses as was recently suggested (Vanhoutte et al., 2013a). Plans are underway to assess patients with MMN and MGUSP over

a much longer period of time. Also, although beyond the scope of the current study, there were differences in demographics and characteristics between the two populations recruited. All cross-cultural patients were Dutch and the longitudinally examined patients came from different countries. Furthermore, some differences were seen between the two groups such as age and gender distribution. These differences may also contribute to differences in response and findings in the current study. However, despite these limitations, the current study is the largest international collaboration between experts in the field of inflammatory neuropathies involving 26 centers in 10 countries. The findings in the current study are quite robust, particularly for patients with GBS and CIDP.

In conclusion, the Jamar dynamometer and the Vigorimeter have equal clinimetric properties. Patients with immune-mediated neuropathies prefer grip strength assessment using the Vigorimeter. Therefore, the Vigorimeter is the recommended instrument for grip strength assessment in future studies involving patients with GBS and CIDP. Additional studies might improve the observed poor responsiveness in patients with MGUSP and MMN.

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Appendix

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